

Systematic elimination of local gravitational models on the SPARC 175 sample: Evidence for an irreducible galaxy-specific degree of freedom

Morten Magnusson
Symbiose Research, Sandnes, Norway
ORCID: [0009-0002-4860-5095](https://orcid.org/0009-0002-4860-5095)
morten@magnusson.as

April 2026 | Figshare preprint
DOI: [10.6084/m9.figshare.32029704.v1](https://doi.org/10.6084/m9.figshare.32029704.v1)

Abstract. We present a systematic elimination study of galaxy rotation curve models using the SPARC 175 sample. We test a broad class of baryon-based models of the form $V_{\text{obs}}^2 = V_{\text{bar}}^2 \cdot \mu(\cdot)$, including exponential (Bose–Einstein), power-law (screening), and regularised forms, as well as extensions incorporating local gradients (∇g), surface mass density (Σ), and angular velocity (ω). All models with purely global parameters fail in low-acceleration systems (FLOW regime), achieving win rates $\approx 27\%$ against NFW and median $\chi^2/\text{dof} \sim 10$, independent of functional form.

We introduce a minimal hybrid model with a single galaxy-specific amplitude, $V_{\text{obs}}^2 = V_{\text{bar}}^2 \cdot \mu(g) + A_{\text{gal}} f(r)$. This model achieves FLOW win rates of 72.6% and median $\chi^2/\text{dof} = 2.71$, comparable to NFW halo fits (2.64) but with half the number of free parameters per galaxy (one versus two).

We demonstrate that A_{gal} is a stable, intrinsic galaxy property: it is consistent across radial subsets ($\rho = 0.74$), robust under leave-one-out tests (median CV = 12%), and exhibits stable sign (76% agreement). Its amplitude correlates with dynamical scale ($|A|$ vs V_{max}^2 , $\rho = 0.53$), but neither its magnitude nor sign can be predicted from baryonic observables: A_{gal} is orthogonal to all tested baryonic properties ($R^2 = 0.02$), indicating that it encodes independent dynamical information.

We conclude that galaxy rotation curves require at least one additional degree of freedom beyond the baryonic mass distribution. The failure of all purely local multiplicative models indicates that the missing physics is not in the functional form of μ , but in an unobserved galaxy-specific parameter.

Keywords: galaxy rotation curves, modified gravity, radial acceleration relation, dark matter, model selection, SPARC

Introduction

The radial acceleration relation (RAR) demonstrates a tight empirical correlation between the observed centripetal acceleration g_{obs} and the baryonic gravitational acceleration g_{bar} in disc galaxies [1,2]. This relation is commonly parameterised as $g_{\text{obs}} = g_{\text{bar}} \cdot \mu(g_{\text{bar}})$ for some interpolation function μ , raising the question of whether μ represents a fundamental gravitational law or an effective description of a more complex underlying physics.

In this work we ask a sharper question: *is any model of the form $V^2 = V_{\text{bar}}^2 \cdot \mu(\text{local info})$, with purely global parameters and zero per-galaxy freedom, sufficient to reproduce the rotation curves of all galaxy types?*

We test this exhaustively on the SPARC 175 sample [3], the standard benchmark for rotation curve studies. We examine exponential, power-law, and regularised μ -forms; extensions to gradient-, surface-density-, and frequency-dependent models; and purely additive dark components. Every model is compared against NFW halo fits with baryonic mass [4], which serve as a two-parameter-per-galaxy reference.

Our central finding is negative: no local model with global parameters reproduces low-acceleration rotation curves. However, adding a single galaxy-specific amplitude parameter is both necessary and sufficient, matching NFW quality with half the parametric freedom.

Data and methods

Sample

We use the SPARC 175 rotation curve dataset [3], comprising 3391 radial data points across 175 galaxies spanning three decades in luminosity and surface brightness. Each data point provides the observed rotation velocity V_{obs} , measurement uncertainty σ_V (floored at 1 km s^{-1}), and decomposed baryonic contributions V_{gas} , V_{disk} , V_{bul} .

Baryonic model

The baryonic velocity is computed with frozen mass-to-light ratios:

$$V_{\text{bar}}^2 = V_{\text{gas}}^2 + \Upsilon_d V_{\text{disk}}^2 + \Upsilon_b V_{\text{bul}}^2, \quad \Upsilon_d = 0.5, \quad \Upsilon_b = 0.7 \quad [M_{\odot}/L_{\odot}]. \quad (1)$$

These are the standard SPARC values at $3.6 \mu\text{m}$ [3].

Regime classification

Galaxies are classified into three EFC regimes based on normalised likelihood from a prior single-component analysis [5]: **FLOW** (62 galaxies, low acceleration, gas-dominated), **TRANSITION** (86 galaxies), and **LATENT** (27 galaxies, high acceleration, baryon-dominated). This classification is frozen throughout all tests.

Models tested

We test the following model classes:

Multiplicative models ($V^2 = V_{\text{bar}}^2 \cdot \mu$):

$$\text{Bose-Einstein: } \mu_{\text{BE}} = 1 + \frac{1}{\exp(\sqrt{g/a_0}) - 1} \quad (2)$$

$$\text{Screening: } \mu_{\text{scr}} = \left(1 + \frac{g^\dagger}{g}\right)^k, \quad k = 0.415 \text{ (locked)} \quad (3)$$

Regularised BE (three forms): $\mu_A = 1 + 1/(\mathrm{e}^{\sqrt{g/a_0}} - 1 + \varepsilon)$, μ_B with g -floor, and μ_C with cubic-root exponent.

Gradient-dependent: $\mu = \mu_{\mathrm{scr}} \cdot [1 + (L_g/L_0)^\beta]^{-1}$ where $L_g = g/|dg/dr|$.

Surface-density-dependent: $\mu = (1 + g^\dagger/g)^{k_{\mathrm{eff}}(\Sigma)}$ with $k_{\mathrm{eff}} = k_0(1 + k_\Sigma \tanh[(\log \Sigma - \log \Sigma_0)/\delta])$.

Additive models ($V^2 = V_{\mathrm{bar}}^2 + V_{\mathrm{extra}}^2$): isothermal (V_0^2), cored ($V_0^2 r^2/(r^2 + r_c^2)$), and logarithmic ($V_0^2 \ln(1 + r/r_s)$).

Hybrid model (1 parameter per galaxy):

$$V^2 = V_{\mathrm{bar}}^2 \cdot \mu_{\mathrm{scr}}(g; g^\dagger) + A_{\mathrm{gal}} \frac{r}{r + r_0}, \quad (4)$$

with $g^\dagger$ and r_0 global, A_{gal} free per galaxy.

Reference: NFW halo + baryons (V_{200} , c free per galaxy) [4].

Fitting and comparison

All global parameters are optimised by minimising total χ^2 across 175 galaxies. Per-galaxy parameters (NFW, hybrid) are fitted individually. Model comparison uses $\Delta\mathrm{AIC} = \mathrm{AIC}_{\mathrm{model}} - \mathrm{AIC}_{\mathrm{NFW}}$, with $|\Delta\mathrm{AIC}| > 10$ as decisive.

Control tests

Shuffle test: Permuting the structural variable (gradient or Σ) between galaxies to verify genuine information use. **Numerics check:** Savitzky–Golay smoothing, uniform rebinning, and noise injection to verify results are physical, not artefacts.

Results: the elimination chain

Table 1 summarises the complete elimination sequence.

Table 1: Systematic elimination of rotation curve model classes. All models use frozen Υ and zero per-galaxy parameters unless noted. Win rate is the fraction of galaxies where the model is preferred over NFW+baryons ($|\Delta\mathrm{AIC}| > 10$; ties counted as wins). NFW uses 2 free parameters per galaxy.

Model class	Global params	FLOW win%	LATENT win%	Status
$\mu_{\mathrm{BE}}(g)$, Eq. 2	1 (a_0)	27.4	77.8	Falsified
$\mu_{\mathrm{scr}}(g)$, Eq. 3	1 ($g^\dagger$)	27.4	74.1	Falsified
Regularised BE (ε)	2	29.0	74.1	Falsified
Regularised BE (g_{min})	2	27.4	74.1	Falsified
Regularised BE ($\sqrt[3]{\cdot}$)	1	25.8	66.7	Falsified
$\mu(g, \nabla g)$ gradient	3	27.4	74.1	Falsified [†]
$\mu(g, \Sigma)$ surface density	5	27.4	74.1	Spurious [‡]
Additive isothermal	1	3.2	3.7	Failed
Additive cored	2	14.5	77.8	Inferior
Additive logarithmic	2	21.0	77.8	Inferior
Hybrid $\mu + A_{\mathrm{gal}}$	1 global + 1/gal	72.6	88.9	Matches NFW

[†]Shuffle test $p = 0.36$: gradient information not used.

[‡]Shuffle test $p < 0.001$ but FLOW unchanged; $\Sigma \equiv g$ (collinear, partial $r = 0.00$).

Exponential class (Bose–Einstein)

The BE form (Eq. 2) yields optimal $a_0 = 1.21 \times 10^{-10} \text{ m s}^{-2}$, matching the MOND acceleration scale to within 1% ($a_0/a_{0,\text{MOND}} = 1.009$). Despite this, FLOW galaxies achieve only 27.4% win rate with median $\chi^2/\text{dof} = 12.3$. The BE form diverges as $\mu \sim 1/\sqrt{g/a_0}$ at low g , suggesting divergence as the failure mode.

Regularised forms

Three regularisations were tested to isolate whether the failure is asymptotic (divergence) or structural (curvature). Form A ($\varepsilon = 0.055$, $\mu_{\text{max}} = 19.3$) reduces total χ^2 by 7% but FLOW win rate improves by only 1.6 pp (27.4% $\rightarrow$ 29.0%). Forms B and C perform similarly or worse. The optimiser for Form A found $\varepsilon > 0$, confirming the divergence is suboptimal, but the FLOW improvement is negligible.

Conclusion: the failure is not in the asymptotic behaviour but in the functional shape of $\mu(g)$ across the full acceleration range.

Power-law class (screening)

Switching from exponential to power-law structure (Eq. 3) with locked $k = 0.415$ yields $g^\dagger = 2.66 \times 10^{-10} \text{ m s}^{-2}$ and total χ^2 improvement of 4%, but FLOW win rate is unchanged at 27.4%. Freeing k finds $k = 0.380$, $g^\dagger = 3.44 \times 10^{-10}$ with FLOW at 29.0%. The two functional classes—exponential and power-law—are indistinguishable in FLOW.

Gradient extension

Adding a gradient-length damping factor $f(L_g/L_0)$ with Savitzky–Golay-smoothed derivatives gives the optimiser full access to local field structure. The optimiser drives $L_0 \rightarrow 10,000 \text{ kpc}$ (upper bound), effectively switching the gradient term off. Shuffle test: $p = 0.36$. The gradient carries no predictive information for rotation curves.

Surface density and angular velocity

A Σ -modulated effective k (five global parameters) achieves χ^2 reduction of 4.8% and passes the shuffle test ($p < 0.001$), confirming that Σ information is genuinely used. However, FLOW win rate remains at 27.4%. Partial correlation analysis reveals $\Sigma \equiv g$ (partial $r = 0.00$ when controlling for g): the apparent signal is collinear with the acceleration already in the model. Adding angular velocity $\omega = V_{\text{bar}}/r$ provides no additional predictive power (partial $r = 0.12$, $p = 0.11$).

Additive models

Universal additive dark components with global parameters perform *worse* than multiplicative models. A constant velocity floor ($V_0 = 69.9 \text{ km s}^{-1}$) achieves only 3.2% FLOW win rate. Cored and logarithmic profiles improve to 14.5% and 21.0% respectively, but remain inferior to screening.

Results: the hybrid model

Adding a single free amplitude A_{gal} per galaxy (Eq. 4) transforms the results. Table 2 shows the regime breakdown.

Table 2: Regime-specific performance: screening (0 params/galaxy), hybrid (1 param/galaxy), and NFW (2 params/galaxy).

Regime	Screening win%	Hybrid win%	Med. χ^2/dof (Hybrid)
FLOW (62)	27.4	72.6	2.85
TRANSITION (86)	23.3	65.1	1.71
LATENT (27)	74.1	88.9	6.88
All (175)	32.6	71.4	2.71

The global parameters are $g^\dagger = 2.66 \times 10^{-10} \text{ m s}^{-2}$ and $r_0 = 2.37 \text{ kpc}$. The hybrid model achieves median $\chi^2/\text{dof} = 2.71$ across all galaxies, compared to 2.64 for NFW—essentially identical quality with half the per-galaxy parametric freedom.

Properties of A_{gal}

Correlation analysis

Table 3 shows that A_{gal} is effectively uncorrelated with all tested baryonic observables. Multiple regression yields $R^2 = 0.02$: baryonic properties explain only 2% of the variance in A_{gal} .

Table 3: Spearman rank correlations between A_{gal} and galaxy observables.

Observable	ρ	p -value
$\log M_{\text{bar}}$	−0.144	0.057
$\log V_{\text{max}}$	+0.054	0.478
$\log R_{\text{last}}$	−0.032	0.676
$\log \Sigma_{\text{mean}}$	−0.136	0.074
f_{gas}	+0.124	0.101
$\log \bar{g}$	−0.136	0.074
$\log \bar{\rho}$	−0.158	0.037
$ A_{\text{gal}} \text{ vs } V_{\text{max}}^2$	+0.533	3.3×10^{-14}

The single strong correlation is between $|A_{\text{gal}}|$ and V_{max}^2 ($\rho = 0.53$): the *amplitude* of the correction scales with dynamical mass, but the *sign* is unpredictable.

Stability

We split each galaxy at its radial midpoint and fit A_{gal} independently on the inner and outer halves (143 galaxies with ≥ 8 points). Results (Table 4) confirm that A_{gal} is a stable galaxy property, not a fitting artefact.

Regime dependence

A_{gal} shows systematic regime dependence: FLOW galaxies have median $A = -826 \text{ (km/s)}^2$ (screening over-predicts $\rightarrow$ correction is negative), while LATENT galaxies have median $A = +330 \text{ (km/s)}^2$ (baryons under-predict $\rightarrow$ correction is positive).

Table 4: Stability and robustness of A_{gal} .

Metric	Value
Inner–outer Spearman ρ	0.741 ($p = 4 \times 10^{-26}$)
Inner–outer Pearson r	0.759 ($p = 5 \times 10^{-28}$)
Sign agreement	76.2%
LOO median CV	11.9%
LOO median sign-flip rate	0.0%

Sign determinants

No tested observable predicts the sign of A_{gal} : rotation curve outer slope ($\rho = -0.001$), concentration ($\rho = 0.012$), and gas fraction ($\rho = 0.124$) are all consistent with noise. The sign is not determined by any tested observable.

Relation to NFW parameter space

A natural concern is whether A_{gal} is merely a reparametrisation of the NFW halo parameters (V_{200}, c). We test this by fitting NFW+baryons to all 175 galaxies and correlating the best-fit (V_{200}, c) with A_{gal} .

NFW parameters explain only 6.3% of the variance in A_{gal} ($R^2 = 0.063$ for multiple regression on $\log V_{200}$ and $\log c$). The signed correlations are weak: A_{gal} vs V_{200} gives $\rho = +0.23$ ($p = 0.003$), and A_{gal} vs c gives $\rho = +0.35$ ($p = 2 \times 10^{-6}$). The amplitude $|A_{\text{gal}}|$ correlates moderately with V_{200} ($\rho = 0.38$), but weakly with c ($\rho = -0.12$, $p = 0.12$).

Critically, the regime structure differs: within FLOW galaxies, A_{gal} correlates strongly with concentration ($\rho = 0.63$, $p = 5 \times 10^{-8}$) but not with V_{200} ($\rho = 0.14$, $p = 0.26$). In LATENT galaxies, A_{gal} is uncorrelated with both ($\rho \approx 0$). This regime-dependent mapping is inconsistent with a simple reparametrisation.

The residual comparison is also informative. Across all 3391 data points, the hybrid model achieves lower mean absolute fractional residuals than NFW (0.096 vs 0.126), despite using half the per-galaxy parameters. In FLOW specifically, the hybrid residuals are 0.089 vs NFW’s 0.105.

We conclude that A_{gal} is not a compression of the NFW (V_{200}, c) pair. It encodes independent dynamical information, and the screening function $\mu(g)$ absorbs the role that NFW’s concentration parameter c would otherwise play.

Numerics verification

To exclude data-processing artefacts, we repeated the screening analysis under three perturbations:

- **Smoothing:** Savitzky–Golay filter on baryonic components. FLOW χ^2/dof changes by 0.03 (0.3%).
- **Rebinning:** Interpolation to uniform 20-point r -grids. FLOW χ^2/dof changes by 0.52 (4.5%).
- **Noise injection:** 1% Gaussian noise on V_{obs} shifts FLOW χ^2/dof by 3.2% (std).

The FLOW/LATENT χ^2 ratio is stable at 0.31 ± 0.01 under all perturbations. The regime separation is invariant to data processing.

Discussion

Implications for the RAR

The standard RAR posits that g_{obs} is a universal function of g_{bar} [1,2]. Our results show that this is an incomplete description: two galaxies with identical g_{bar} profiles can require different rotation curves, parameterised by A_{gal} . The RAR is not a strict one-parameter relation in the low-acceleration regime.

Relation to MOND and Λ CDM

In MOND, the interpolation function $\nu(g_{\text{bar}}/a_0)$ is assumed universal. Our elimination chain falsifies this assumption for any functional form of ν , not just a specific one. In Λ CDM, the dark matter halo provides two per-galaxy parameters (V_{200} , c), which our results show is one more than necessary.

The hybrid model (Eq. 4) can be interpreted either as a screening law with an additive dark component, or as a modified gravitational law requiring an environmental amplitude. The data do not distinguish these interpretations.

What A_{gal} might represent

A_{gal} has units of $(\text{km/s})^2$ and its amplitude scales with V_{max}^2 ($\rho = 0.53$). Physically, it may represent:

- The amplitude of a dark matter halo (analogous to NFW's V_{200}^2), reduced to a single effective parameter.
- An environmental coupling constant in modified gravity, reflecting formation history or tidal environment.
- A non-local gravitational contribution from the global mass distribution, not captured by local g_{bar} .

We do not adjudicate between these possibilities; we establish only that the parameter is necessary, stable, and irreducible.

Robustness of the MOND scale

Across all models tested, the critical acceleration scale converges to $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$, consistent with the MOND value. This convergence is model-independent: it appears in both exponential and power-law functional forms. The MOND scale is robust even though the MOND programme (single universal μ) is not.

Reproducibility

All analysis code is publicly available. Frozen parameters: $\Upsilon_d = 0.5$, $\Upsilon_b = 0.7$, $k = 0.415$ (locked), $\sigma_{\text{min}} = 1 \text{ km s}^{-1}$, seed = 42.

Data: SPARC 175 rotation curves from Lelli, McGaugh & Schombert (2016) [3], with regime classification from [5].

Code: 12 Python scripts in the repository `scripts/sparc175_*.py`, with results in `scripts/sparc175_*_res`

Summary: `scripts/SPARC175_ELIMINATION_SUMMARY.md`.

References

- [1] McGaugh, S. S., Lelli, F., Schombert, J. M. (2016). Radial acceleration relation in rotationally supported galaxies. *Phys. Rev. Lett.* 117, 201101. [doi:10.1103/PhysRevLett.117.201101](https://doi.org/10.1103/PhysRevLett.117.201101)

- [2] Lelli, F., McGaugh, S. S., Schombert, J. M., Pawlowski, M. S. (2017). One law to rule them all: the radial acceleration relation of galaxies. *ApJ* 836, 152. [doi:10.3847/1538-4357/836/2/152](https://doi.org/10.3847/1538-4357/836/2/152)
- [3] Lelli, F., McGaugh, S. S., Schombert, J. M. (2016). SPARC: mass models for 175 disk galaxies with Spitzer photometry and accurate rotation curves. *AJ* 152, 157. [doi:10.3847/0004-6256/152/6/157](https://doi.org/10.3847/0004-6256/152/6/157)
- [4] Navarro, J. F., Frenk, C. S., White, S. D. M. (1997). A universal density profile from hierarchical clustering. *ApJ* 490, 493. [doi:10.1086/304888](https://doi.org/10.1086/304888)
- [5] Magnusson, M. (2026). Kill-Test v6 Universality—Extension of the EFC vs Λ CDM Kill-Test to the Full SPARC 175 Sample. Figshare. [doi:10.6084/m9.figshare.31986762](https://doi.org/10.6084/m9.figshare.31986762)
- [6] Magnusson, M. (2026). GR Recovery in Energy-Flow Cosmology via Density Saturation: Quantitative PPN Analysis. Figshare. [doi:10.6084/m9.figshare.31244827](https://doi.org/10.6084/m9.figshare.31244827)
- [7] Magnusson, M. (2026). Covariant EFT for Entropy-Driven Gravitational Modification: Construction, Constraints, and an Identified Microphysical Gap. Figshare. [doi:10.6084/m9.figshare.31878334](https://doi.org/10.6084/m9.figshare.31878334)
- [8] Milgrom, M. (1983). A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis. *ApJ* 270, 365. [doi:10.1086/161130](https://doi.org/10.1086/161130)
- [9] Li, P., Lelli, F., McGaugh, S. S., Schombert, J. M. (2018). Fitting the radial acceleration relation to individual SPARC galaxies. *A&A* 615, A3. [doi:10.1051/0004-6361/201732547](https://doi.org/10.1051/0004-6361/201732547)
- [10] Magnusson, M. (2026). Component-weighted bias in rotation curve model selection. Figshare.
- [11] Akaike, H. (1974). A new look at the statistical model identification. *IEEE Trans. Autom. Control* 19, 716. [doi:10.1109/TAC.1974.1100705](https://doi.org/10.1109/TAC.1974.1100705)